

L05 – Chihuahua or Muffin with CNN Reflection

This lab reinforced what we have been learning about convolutional neural networks (CNNs) and image classification. In the previous Chihuahua or Muffin workshop, we used a traditional neural network. This time, we used a CNN, which is designed specifically for images. One thing this lab reinforced for me is that a CNN does not treat an image like one long list of numbers. Instead, it uses convolutional and pooling layers to recognize patterns like edges, shapes, and textures before making a prediction. Because it keeps the spatial information in the image, a CNN is usually better at image classification than a traditional neural network.

After running the notebook, I wanted to see if I could improve the model by changing some of the training settings. Instead of changing several things at once, I only changed one setting at a time so I could clearly see how each change affected the results.

The first thing I experimented with was the number of epochs. I played with the training by using 5 epochs, then 10 epochs, and finally 15 epochs. With 5 epochs, the model trained faster, but the final validation accuracy was only 93.33%. When I increased the training to 10 epochs, the validation accuracy improved to 96.67%, and the model reached 100% validation accuracy during training. I wanted to see if training even longer would improve the model, so I increased it to 15 epochs. Even though the model again reached 100% validation accuracy during training, the final validation accuracy stayed at 96.67%. That showed me that simply training longer does not always improve the final model, so I decided that 10 epochs gave the best balance between training time and performance.

Next, I experimented with the learning rate. The notebook used a learning rate of 0.001, so I lowered it to 0.0005 to see if the model would learn differently. This ended up producing the best overall results because the model finished with 100% validation accuracy. Finally, I experimented with the CNN architecture by changing the dropout layer from 0.5 to 0.3. The model still performed very well and reached 100% validation accuracy during training, but the final validation accuracy finished at 96.67%. These experiments showed me that even small changes to the training settings can affect how well a model performs.

Compared to the traditional neural network from the previous workshop, I thought the CNN handled image classification much better. The traditional neural network flattened the image before making predictions, while the CNN automatically learned important image features through its convolutional layers. The CNN also took longer to train because it had more layers and many more parameters, but the improved performance made the extra training time worth it. Reinforcing the differences between these two models helped me better understand why CNNs are the standard for most computer vision problems.

One challenge I had during this lab happened before I even started. I could not find the notebook because it was missing from the GitHub repository. I emailed my instructor about the problem because I wanted to finish my assignments before leaving for my trip. While waiting for a response, one of my classmates shared the notebook with me so I could continue working. Another challenge was deciding which settings actually worked best. After every change I made, I reran the notebook, compared the new accuracy numbers, and decided whether the change actually improved the model. Changing one setting at a time made it much easier to understand what each experiment was doing.

CNNs have many real-world applications beyond this lab. They are used to help detect diseases from medical images, recognize faces for security systems, inspect products during manufacturing, identify objects for self-driving cars, and organize photos automatically. This lab helped me see how the same type of model used to classify Chihuahuas and muffins can also be used to solve much larger real-world problems.

Even though CNNs are very powerful, there are also ethical concerns. If a model is trained with biased or limited data, it may make unfair or incorrect predictions. Privacy is another concern because many computer vision systems use personal images or facial recognition. AI should also support human decision-making instead of completely replacing it, especially in healthcare, law enforcement, or other situations where mistakes could seriously affect people.

Overall, this lab reinforced how effective CNNs are for image classification. My biggest takeaway was that small changes to the training settings can make a big difference in performance. Instead of assuming one model was the best, I tested different configurations, compared the results, and chose the settings that performed the best. That process helped me better understand how CNNs learn and why experimenting is an important part of machine learning.

References

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